

The tokenised-fund plumbing gap

Why in-kind on-chain issuance and a credibly-neutral NAV are a real, frequent, quantifiable pain — with sources.

1 · The problem, quantified

Tokenised funds have scaled fast, but the **fund machinery** — NAV striking, primary-market creation/redemption, reconciliation — was left in the TradFi back office. The asset moved on-chain; the operations did not.

\$30B+

tokenised RWA on-chain, ~2× since Jan 2025¹

\$15.2B

tokenised US Treasuries, 76 products, May 2026²

8

chains one fund (Franklin BENJI) spans — fragmented issuance³

- ▶ **NAV is manual and slow.** Valuations, accruals and bank reconciliations sit in separate systems; "weeks of work compress into the final days of each reporting cycle."⁴ Fees are "the most complex and error-prone component."⁴
- ▶ **Reconciliation drift erodes trust.** "Units on issue × NAV per unit should equal fund-level NAV... persistent discrepancies indicate unit-ledger drift or unbooked transactions."⁴ A shared ledger removes this class of error entirely.⁵
- ▶ **No trustworthy on-chain NAV.** Without one, a tokenised fund still depends on a single administrator you must trust — reintroducing the counterparty the tokenisation was meant to remove.

2 · Why now — regulation and assets converged in mid-2026

- ▶ **SEC approved IN-KIND creation & redemption for spot BTC/ETH ETFs on July 29, 2025.**⁶ Crypto ETFs were cash-only before; now APs can create/redeem directly in the underlying crypto — the tax-efficient mechanism⁷ — but **no on-chain venue performs it natively.**
- ▶ **DTCC is beginning to tokenise DTC- & Fed-eligible securities on Canton.** DTCC custodies ~\$114T (its total book — the ceiling, not what's live); ran its **first live production trades in July 2026** (a pilot); 50+ firms incl. BlackRock & JPMorgan; DTCC + Euroclear co-chair Canton governance.⁸ *Live tokenised market today: ~\$30B.*¹
- ▶ **The underlyings are live on Canton:** cETH (OnRails), CBTC (BitSafe), USDC (Circle) — the exact components a tokenised basket is created from.

3 · The North Star metric

Not "number of chains" or "number of parties." The pain is **operational friction and time-to-issue**. CrossDesk's value is measured as:

NORTH STAR

Settlement friction removed per creation/redemption: minutes-to-issue & reconciliation cost eliminated, and principal-risk exposure removed (→ 0).

DIMENSION	TODAY (OFF-CHAIN FUND OPS)	CROSSDESK (ON-LEDGER)
NAV strike	Manual, spreadsheet, days ⁴	Committee-attested each session, on-ledger
Create / redeem	Back-office in-kind, T+1/T+2	Atomic, single transaction, instant finality
Reconciliation	Daily across ledger/custodian/broker ⁴	None — the ledger is the register ⁵
Principal risk	Present between legs	Zero (all-or-nothing DvP)
NAV trust	Single administrator	K-of-N committee, provable from signatures

4 · Is the pain real, meaningful, and frequent? (the judge's test)

- ▶ **Real:** fund administration is a dedicated multi-billion-dollar industry precisely because NAV + reconciliation + primary issuance are hard.^{4,5}
- ▶ **Meaningful:** the in-kind mechanism is why ETFs are more tax-efficient than mutual funds — in-kind swaps don't trigger capital gains.⁷ The SEC extended it to crypto in 2025⁶ precisely because it matters.
- ▶ **Frequent:** creation/redemption and NAV striking happen **every trading day, per fund**. As \$114T of securities tokenise on Canton,⁸ the frequency scales with the whole market.

5 · The competitive gap

Canton has OTC desks and continuous orderbooks. It has token issuers (OnRails, BitSafe, Circle) and, soon, tokenised securities (DTCC). It does **not** have an on-chain **primary-market fund-issuance layer with a decentralised NAV**. That primitive is unoccupied — and it's what CrossDesk builds.

Sources

- 1 CryptoSlate / RWA.xyz, tokenised RWA ~\$30B, 2026.
- 2 RWA.xyz — tokenised US Treasuries \$15.2B / 76 products, May 2026.
- 3 InvestmentNews; eco.com — BUIDL ~\$2.1–2.5B, BENJI ~\$828M across 8 chains, 2026.
- 4 getcaruso.com "What is NAV in Fund Administration"; navquant.com — manual NAV, fee complexity, reconciliation drift.
- 5 Chainlink, "Blockchain in Fund Administration" — unified ledger removes counterparty reconciliation.
- 6 SEC.gov (Uyeda statement, 2025-07-29); CoinDesk; Dechert; Morrison Foerster — in-kind creation/redemption approved for spot BTC/ETH ETPs.
- 7 State Street; ETF.com; Britannica Money; Schwab — in-kind swaps are tax-efficient (no capital-gains event).
- 8 DTCC.com (2025-12-17); CoinDesk (2025-12-17); canton.network; @CantonNetwork — DTCC tokenising DTC/Fed-eligible securities on Canton, \$114T custodied, live production trades July 2026, DTCC+Euroclear co-chair governance.

CrossDesk · Sebastian Borjas · HackCanton S2 · Track: Investment Infrastructure — Funds & Governance · Build settles cETH & CBTC on Canton Devnet · repo: github.com/superbigroach/canton-dvp-settlement-desk